

CSE 461 : Introduction to Robotics

Lab Report

Lab No: 02
Lab Task: Measuring distance using ultrasonic sensor.

Group Information

Lab Section: 04
Group Number: 01

Group Members:

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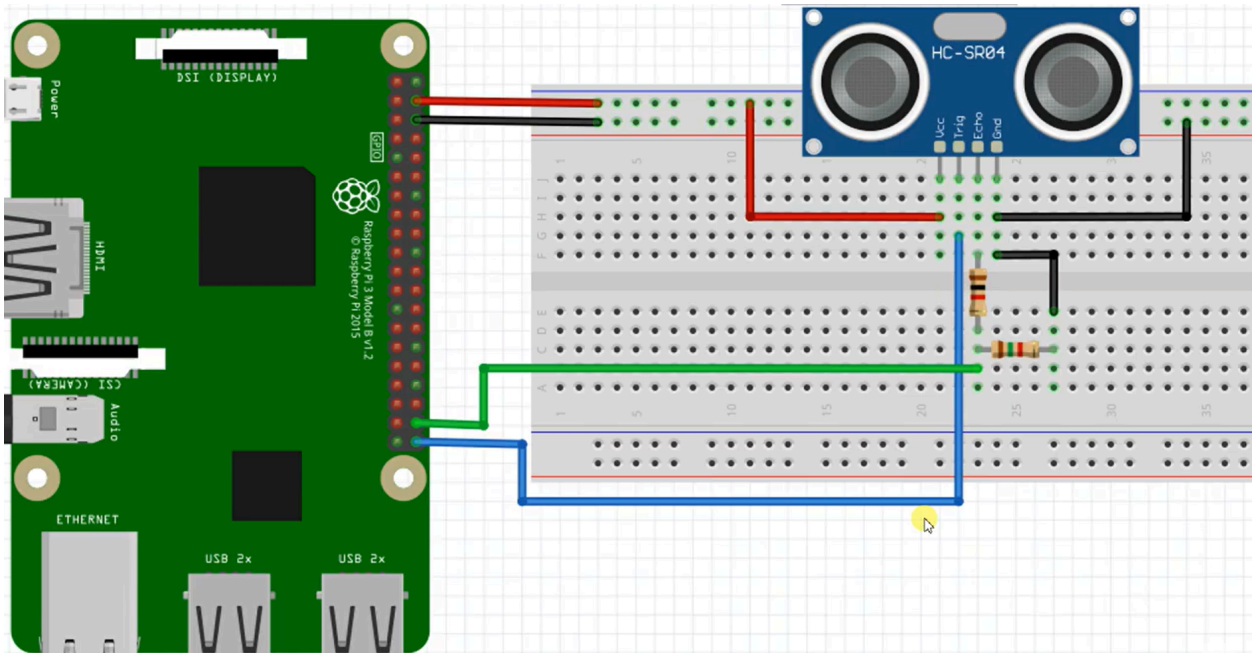
Task Description:

The objective of the experiment is to get to know about ultrasonic sensors and its functionalities. We know that ultrasonic sensors are used to measure distance. The sensors send out sound waves and calculate the distance by measuring the time it takes for the waves to bounce back. The aim of this experiment is to measure distance by grasping the concept of how ultrasonic sensors are installed and used along with a block of code and a few other hardware components, notably Raspberry Pi.

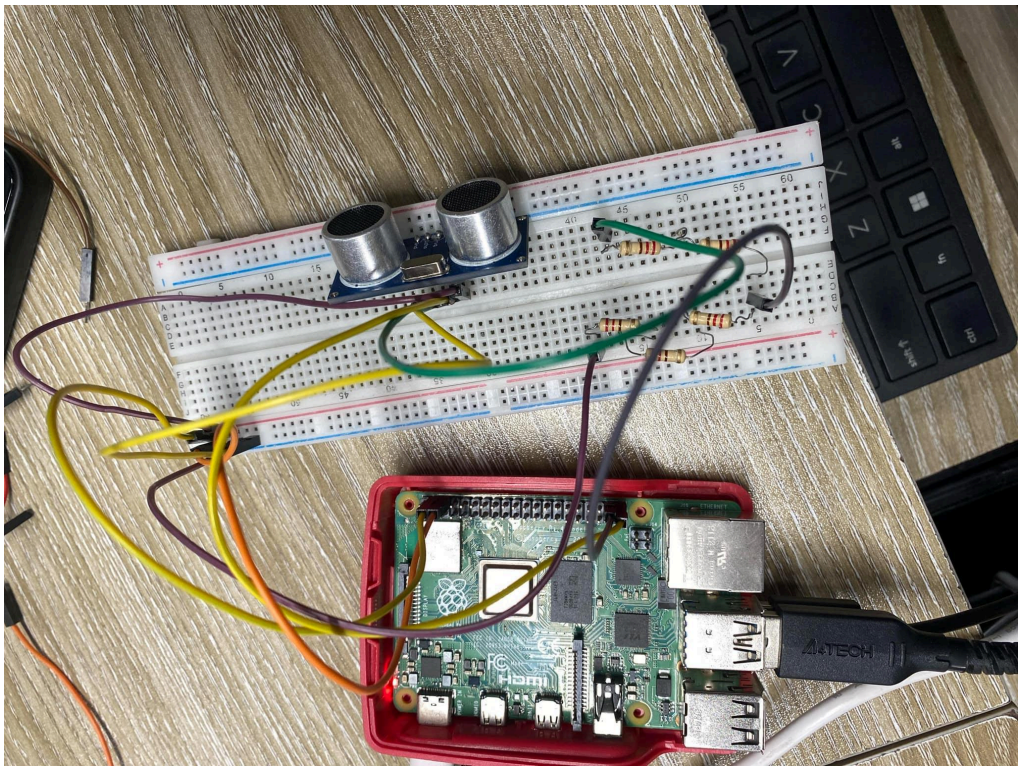
Components Used:

- Raspberry Pi 4
- Ultrasonic sensor (HC-SR04)
- Breadboard
- Jumper Wires
- 1k and 1.5k resistors
- MicroSD Card
- USB Cable
- Monitor, Keyboard, and Mouse.

Circuit Diagram:



Circuit Setup:



Code:

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BCM)

TRIG = 21
ECHO = 20

GPIO.setup(TRIG,GPIO.OUT)
GPIO.setup(ECHO,GPIO.IN)

def distance():
        GPIO.output(TRIG, False)
        time.sleep(0.5)
        GPIO.output(TRIG, True)
        time.sleep(0.00001)
        GPIO.output(TRIG, False)
```

```
        pulse_start = time.time()

        while GPIO.input(ECHO)==0:
                pulse_start = time.time()

        while GPIO.input(ECHO)==1:
                pulse_end = time.time()

        pulse_duration = pulse_end - pulse_start

        distance = pulse_duration * 17150

        distance = round(distance, 2)

        return distance

print(distance())

GPIO.cleanup()
```

Discussion / Q&A:

The experiment was a thorough overview of what an ultrasonic sensor is, what it does, how it works and how to use it to measure distances. Moreover, this experiment also helped us understand the use and implementation of program code along with Raspberry Pi and other components to successfully measure distances using an ultrasonic sensor. Following the connection guidelines, we were successfully able to carry out the task.

1) Why are the resistors used?
Answer: The Ground pins of a Raspberry Pi are not designed to tolerate high voltage connections. These ground pins are used to provide a common electrical ground connection to the components. Applying high amounts of voltage may damage the Raspberry Pi along with the components in connection with it. So, resistors are used to divide the high voltage power input (5V in this case) into reduced voltage amounts.